12.14 Mixed Strategies

WORKSHEET

The question

STEP I 1989, Question 16

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s1-q16

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself Write down a table of B's winnings for each pair of guesses (B's guess down the side, A's across the top).

STAGE 1 Against a fixed guess

Covers: Part (i)

Ask yourself Make a table of B's winnings for each pair of guesses. If A always says the same number, which entry should B pick?

STAGE 2 Against random guessing

Covers: Part (ii)

Ask yourself If A guesses each number with probability $\frac{1}{3}$, what is B's expected gain when B says j? Average along a row of the table.

STAGE 3 The general case

Covers: Part (iii)

Pause and try Collect B's expected winnings by A's guess, giving coefficients of a_1, a_2, a_3. Can you choose the b_j to make all three zero? Check whether the three equations are independent.

What's really going on

The minimax theorem

In a two-player *zero-sum* game, given by a payoff matrix M, B choosing probabilities $\mathbf{b}$ and A choosing $\mathbf{a}$, B's expected winnings are $\mathbf{b}^{\mathsf{T}} M \mathbf{a}$. Von Neumann's minimax theorem says

$$\max_{\mathbf{b}} \min_{\mathbf{a}} \mathbf{b}^{\mathsf{T}} M \mathbf{a} = \min_{\mathbf{a}} \max_{\mathbf{b}} \mathbf{b}^{\mathsf{T}} M \mathbf{a}.$$

In words: the amount B can guarantee by announcing a randomised strategy in advance equals the amount A can hold B to by announcing his. This common value is the *value* of the game. Here, by antisymmetry, the value is 0, and the strategy $(\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$ is optimal for both players. The theorem is proved using linear algebra and convexity (it is equivalent to the duality theorem of linear programming), and its generalisation to

many players, the *Nash equilibrium*, won John Nash the Nobel Prize in economics.

Why randomise?

Randomness here is not about luck but about information. If B's choice can be predicted, A can exploit it; by randomising with the right probabilities, B makes his choice unpredictable *and* ensures that no prediction would help. Real examples include penalty kicks in football, where studies of professional players have found kickers and goalkeepers randomising between left and right in proportions close to the game-theoretic optimum, and tennis serves.

Von Neumann's 1928 paper

John von Neumann proved the minimax theorem in 1928, in a paper "On the theory of games of strategy", motivated by poker, where bluffing is a form of randomisation. In 1944 he and the economist Oskar Morgenstern published *Theory of Games and Economic Behavior*, which founded game theory as a subject. Von Neumann later recalled that he had regarded the minimax theorem as the key result, without which there would be no theory of games worth publishing.

Try this next

1. **A's best reply.** If B uses $(\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$, show that every strategy of A gives expected winnings 0. If instead B uses $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$, find A's best reply.

2. **Rock, paper, scissors.** Write the payoff matrix for rock–paper–scissors and find the optimal strategy. Why is it uniform here but not in the question?

3. **Uniqueness.** Show that $(\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$ is the *only* strategy for B that guarantees not losing. (*Hint:* it must make X, Y, Z all non-negative; use $X + 2Y + Z = 0$.)

4. **A modified game.** If the penalty for differing by 2 is £3 instead of £2, find the optimal strategy.

Further reading

- **Ken Binmore, *Playing for Real*** (Oxford University Press). A thorough but friendly introduction to game theory.
- **Avinash Dixit & Barry Nalebuff, *The Art of Strategy*** (Norton). Game theory with examples from sport and business.
- **William Poundstone, *Prisoner's Dilemma*** (Anchor). Von Neumann, game theory and the Cold War.

The full guided solution is in Section 12.14:
https://beyondstep.keithmcnulty.org/ch12/mixed-strategies.html